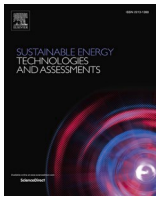
Sustainable Energy Technologies and Assessments 93 (2026) 105278



Contents lists available at ScienceDirect

Sustainable Energy Technologies and Assessments

journal homepage: www.elsevier.com/locate/seta



Techno-economic assessment of hydrogen-based direct reduction for India's steel sector: comparative analysis with natural gas DRI and policy enablers

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ARTICLE INFO

Keywords:
Techno-economic analysis
Iron and steel
Natural gas
Hydrogen
Process engineering
Decarbonization

ABSTRACT

The iron and steel sector contributes nearly 7% of the global energy-related CO_2 emissions. In India, this challenge is amplified due to high production of coal based direct reduced iron (DRI) (~80% share). This research evaluates the techno-economic feasibility of hydrogen-based direct reduction (H_2–DRI) for India's steel industry and benchmarks its results against natural gas-based DRI (NG–DRI), under realistic market conditions. A 1 million tons per annum (MTPA) H_2–DRI plant and NG-DRI plant is modeled using ASPEN Plus v14, and economic estimates are developed with sensitivity to renewable electricity tariffs, load factor, and policy levers such as carbon pricing and green steel premiums. By comparing DRI based on natural gas, hydrogen and coal the impact of capital expenditure, operational expenditure and overall economics in terms of levelized cost of iron production is evaluated. Results indicate that demand side dynamics like green steel premium and carbon pricing are important factors towards economic viability of H_2 based DRI.

Introduction

The world encounters the challenge posed by continued greenhouse gas emissions which are projected to increase global warming, with scenarios indicating a likely rise in the global average temperature this century to 1.5°C above pre-industrial levels [1]. In 2024, global energy related CO_2 emissions were approximately 37.8 gigatons, showcasing the magnitude of efforts required in achieving net-zero emission targets [2]. The iron and steel sector remains one of the most carbon intensive sectors, emitting ~ 2.6 Gt CO_2 annually, equivalent of ~ 7% of global energy system emissions [3]. In this scenario, conventional steel manufacturing, an important activity for industrial growth emits on an average of 2.18 t CO_2 per ton (t) of crude steel, with energy consumption of ~ 21 GJ/t. The share of renewable energy in iron and steel production is below 2%, and several efforts are in place to further increase its penetration [4].

It is widely accepted that decarbonization of the iron and steel industry requires a strategic approach which addresses both process efficiency and fuel substitution. Energy efficiency improvements, such as advanced furnace technologies and waste heat recovery, have the promise to deliver immediate reductions in energy intensity [5]. Increasing scrap utilization through Electric Arc Furnace (EAF) routes further lowers emissions by reducing reliance on virgin iron ore [5]. Transitioning from coal-based rotary kiln DRI to Natural Gas-based DRI (NG-DRI) offers a pragmatic intermediate step, cutting CO_2 emissions by nearly 50–70% compared to coal-based routes while leveraging existing shaft furnace infrastructure, specifically in developing countries like India [6].

In the long term, Hydrogen-based Direct Reduction (H_2-DRI) represents a near-zero carbon pathway, with lifecycle emissions below 0.1 t CO_2 per ton of steel when powered by renewable electricity [7]. These measures can further be augmented by Carbon Capture, Utilization, and Storage (CCUS) through which residual emissions in Blast Furnace-Basic Oxygen Furnace (BF-BOF) and transitional DRI processes can be further

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https://doi.org/10.1016/j.seta.2026.105278
Received 2 March 2026; Received in revised form 29 July 2026; Accepted 31 July 2026
Available online 12 August 2026
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minimized [8].

Replacing coal with hydrogen in shaft furnaces is an important step towards achieving the goal of net-zero ironmaking [9]. When hydrogen is produced via renewable-powered electrolysis and process heat is fully electrified, CO_2 emissions can be significantly reduced. Hydrogen based iron-oxide reduction also yields ultra-pure iron with no dissolved carbon, differentiating it from conventional pig iron [10,11]. The technical feasibility of this approach has been further demonstrated by global projects such as MIDREX-H_2 and HYBRIT [12–14].

However, there are major economic barriers persisting in deployment of hydrogen-based steel production on a commercial scale:

- High initial investment for green hydrogen production despite projected cost declines [15].
- Renewable electricity cost and intermittency, requiring high load factors for competitive hydrogen [8].
- Development of Hybrid power purchase agreements and storage as enablers for firm renewable supply [16,17].
- Policy levers such as carbon pricing, green steel premiums, and capital expenditure (CAPEX) subsidies are essential to bridge the cost gap [18,19]. EU benchmarks RED III, the Innovation Fund, and the EU Hydrogen Bank illustrate how mandates and financial instruments accelerate H_2–DRI adoption [20,21].

India is the world's second-largest steel producer, expected to supply ~ 20% of global steel by 2050 [22]. Its production structure is dominated by coal-based rotary kiln DRI (~80%), making it one of the most carbon-intensive globally [23,24]. Emission intensities of coal-based plants often exceed 2.5 t CO_2/t steel, compared to the global average of ~ 1.8 t CO_2/t [8]. The Indian iron ore and non-coking coal is concentrated in five states (Odisha, Chhattisgarh, Jharkhand, Karnataka, and West Bengal) which results in low capital costs and constitutes 86% of country's sponge iron capacity [25,26].

The Ministry of Steel's 2024 roadmap outlines pathways for green transition, including energy efficiency, renewable integration, and gradual integration of hydrogen based technologies [25]. The recently introduced taxonomy for green steel defines thresholds for CO_2 intensity and sets benchmarks for "percentage greenness" rating system, which helps in creating benchmarking and certification.

The National Green Hydrogen Mission and Ministry of New and Renewable Energy MNRE's 2024 guidelines have sanctioned pilot projects for hydrogen use in DRI and Blast Furnace (BF) injections, backed by ~ US$ 50 million funding for scale up and technical feasibility [27]. Think tanks in India like IRADe highlight the dominance of coal-based DRI, limited domestic gas availability, and the need for hybrid renewable power purchase agreements (PPAs) to ensure high electrolyzer utilization [6].

Recognizing this challenge, India launched the National Green Hydrogen Mission and pilot projects for H_2DRI, alongside Round-The-Clock Power renewable auctions to ensure firm renewable supply to achieve a net-zero target by 2070 [16,27].

This article hence focuses on a techno-economic assessment of hydrogen-based direct reduction (H_2-DRI) for the Indian steel industry and is benchmarked against natural gas-based DRI (NG-DRI), by modeling a 1 MTPA plant under Indian economic conditions.

Background

There is broad consensus that deep steel decarbonization requires a shift away from coal–intensive Blast Furnace-Basic Oxygen Furnace (BF–BOF) towards Natural gas Direct Reduction of Iron-Electric Arc Furnace (NG DRI–EAF) and hydrogen–enabled routes, complemented by higher scrap use and selective carbon capture utilization and sequestration (CCUS). The World Steel Association's Sustainability Indicators 2024 report [5] defines performance metrics that explicitly recognize the growing role of DRI–EAF in the global mix, underscoring the route's relevance in emissions intensity tracking and target–setting. On the technology front, HYBRIT has produced more than 5000 tonnes (t) of hydrogen–reduced iron at pilot scale and submitted its six–year results (2018–2024) before moving to the next industrial implementation phase which serves as evidence that H_2–DRI is beyond laboratory maturity and entering scale–up [28]. Future projections in countries like China envisage adoption of H_2 based DRI technology, highlighting its potential as sustainable low carbon route for steel making and projecting 20% to 30% hydrogen based total steel production by 2060 [29,30].

Several studies have examined the thermodynamics and kinetics of hydrogen-based DRI (H_2-DRI), highlighting its potential as a near-zero carbon solution [31,32]. Process modeling of H_2-DRI approaches vary from open-source Python-based mass and energy flow models [10] to ASPEN Plus simulations for green hydrogen production and integration into DRI plants [33]. Facility-level assessments have quantified energy consumption, greenhouse gas (GHG) emissions, and cost implications for electrically driven H_2-DRI plants using water electrolysis [34–36]. Gurgel et al. [37] demonstrate through multiple scenarios that a CO_2 price greater than US$100 per t is required to reduce sectoral emissions by more than 40 %. It also demonstrates that H_2-DRI will be a dominant pathway for future scenario at a green hydrogen price less than US$1 per kg, and substantial reduction in CO_2 emissions could be possibly achieved by switching fuel from coal to natural gas.

According to the life-cycle assessment conducted by Nduagu et al. (2022) [38], greenhouse gas emissions associated with coal-based DRI production range from 1.4 to 2.0 t CO_2e/t-DRI, while NG-DRI production emits 0.8 to 1.1 t CO_2e/t-DRI. Therefore, the adoption of NG-based DRI offers a significant reduction in life-cycle GHG emissions relative to coal-based DRI pathways. Lifecycle analyses also confirm that green hydrogen-based DRI has the promise to achieve emissions below 0.1 t CO_2 per t steel, positioning it as a near-zero carbon pathway [39].

Recent techno-economic studies emphasize that electricity cost, specific energy consumption or electrolyzer efficiency, and load factor are dominant drivers of hydrogen production economics [33,40]. Sensitivity analyses show that increasing load factor significantly reduces the Levelized Cost of Hydrogen (LCOH), while high renewable tariffs and low utilization rates undermine competitiveness [31]. A comparative study between electrolyzer technology Proton Exchange Membrane (PEM) and Alkaline Water Electrolysis (AWE) demonstrates the trade off, in terms of initial capital investment vs system efficiency [41]. In a detailed analysis Rosner et al.[7] have proposed that using renewable H_2 for both iron ore reduction and process heating can cut direct emissions by ~ 85%, with economic viability around US$1.6–1.7 per kg H_2. Furthermore, they have recommended to consider optimizing top–gas recycle pressure from shaft furnace and H_2 excess ratios as process operation options to improve energy use and costs.

Cost modeling for electrolyzer technologies (proton exchange membrane electrolyzer (PEMEL) and Alkaline Electrolyzer (AEL) predicts substantial CAPEX reductions by 2030, with economies of scale delivering major cost benefits up to ~ 10 MW plant size [42]. Further studies have explored integrated hydrogen utilization pathways in ironmaking, combining process modeling with techno-economic and environmental assessment. Kim et al. (2021) [43] demonstrated that a Blast Furnace integrated with a Solid Oxide Electrolysis Cell (BF–SOEC) integration can achieve significant CO_2 reduction, with economic viability strongly dependent on hydrogen production costs. Kim et al. (2026) [44] highlighted the role of alternative hydrogen pathways such as methane pyrolysis and system-level integration. These insights reinforce the importance of hydrogen supply pathways in techno-economic evaluation of H_2-DRI systems.

For natural gas based DRI to effectively replace coal in India which relies on importing energy, it is imperative that a robust gas infrastructure in terms of re-gasification terminal and pipelines is in place to reach to the end industrial customer. It is also essential that imported liquefied natural gas (LNG) is available at competitive and stable price with low CO_2 footprint to ensure affordability and sustainability for the

end user [45].

Despite efforts globally, techno-economic studies catering to needs of coal-based iron production process which can benefit from decarbonization options brought in by an alternate lower carbon intensive fuel such as hydrogen and natural gas remain limited. This article aims to address the following aspects:

- Industrial-scale modeling (1 MTPA H_2-DRI and NG-DRI) as a decarbonization option under Indian market conditions.
- Comparison of H_2-DRI and NG-DRI economics using actual Indian fuel and electricity prices.
- Impact of renewable intermittency and load factor on H_2-DRI viability.
- Comparison of the LCOP of H_2-DRI across varying renewable electricity prices with the benchmark LCOPs of NG-DRI and coal-based DRI production routes
- Policy levers such as carbon pricing and green steel premiums.

The following sections present the results of simulations for a 1 million tons per annum (MTPA) H_2-DRI plant and a NG-DRI plant in India. The analysis estimates capital expenditure (CAPEX), operating expenditure (OPEX), levelized cost of production (LCOP), and net present value (NPV). It also compares the economic performance of H_2-DRI and NG-DRI pathways and evaluates the sensitivity of results to key parameters, including electricity tariffs, plant load factor, carbon pricing, and green premium assumptions. While these techno-economic drivers are well established in literature, their combined application under India-specific cost structures, electricity conditions, and policy context remains limited, which is addressed in this study.

Methodology: Process modelling

The study adopts a techno-economic framework exemplified in Thomassen et al. [46], which is useful in calculating annual energy consumption and material requirements for a steel plant with a production capacity of 1 MTPA of DRI. A DRI plant of that capacity represents the scale of commercial plants being currently considered in this study. A similar capacity techno-economic study has been conducted for establishing a hydrogen direct reduced iron (H_2-DRI) plant in New Zealand [40]. The electrolyzer capacity and H_2 requirement for 1 MTPA plant are within the achievable scale to make this study more realistic. Large scale green hydrogen projects emerging worldwide could cater for the H_2 demand for the DRI plant [47]. This capacity also aligns with typical NG-DRI plant, ensuring a fair and practical comparison as demonstrated in the supplementary material. For comparative assessment, coal-based DRI is included as a reference baseline using literature-reported data, while H_2–DRI and NG–DRI are developed using Aspen Plus process models within this study. In the second step, detailed economic analysis of the H_2-DRI plant is conducted for Indian conditions to evaluate its economic viability, including sensitivity analysis against CAPEX, OPEX and electricity pricing. Third, a comparative assessment of H_2-DRI and NG-DRI is performed for same capacity plant using a financial model to determine the relative competitiveness of NG-DRI against H_2-DRI. Finally, sensitivity analyses of H_2-DRI are carried out on critical variables such as carbon tax and green steel premium to assess their impact on financial viability.

H_2-DRI process description and modelling

The H_2–DRI process is modeled using a shaft furnace based direct reduction configuration in Aspen Plus under steady-state conditions. The model integrates hydrogen production, gas conditioning, shaft furnace reduction, and recycle loops within a single process framework. An alkaline electrolyzer (AEL) is selected in this study based on its higher technology maturity, longer operational lifetime (60–90 k hours vs. 40–60 k hours for PEM), and reliance on non-critical materials (Ni-based catalysts), which enhances supply chain robustness and cost stability [48]. The techno-economic comparison between alkaline (AEL) and PEM electrolyzers has been demonstrated in Supplementary material (Table S3) which indicates a narrowing CAPEX gap, with AEL (~US$235–535/kW) and PEM (~US$245–575/kW) exhibiting comparable cost ranges [49].

H_2 is supplied to the reduction process after compression, purification, and preheating (~800–1000 °C). The reduction gas, consisting of H_2, reacts with descending iron ore in a shaft furnace in a counter-current configuration. The reduction mechanism is represented using stoichiometric reactor blocks simulating stepwise conversion of iron oxides ($Fe_2O_3(s) \rightarrow Fe_3O_4(s) \rightarrow FeO(s) \rightarrow Fe(s)$), based on literature-reported kinetics and thermodynamic feasibility [32,50,51]. Thermodynamic properties are evaluated using the Peng–Robinson equation of state. A hydrogen excess ratio is maintained to ensure a near-complete reduction to metallic iron and to sustain the thermodynamic driving force.

A closed-loop configuration is adopted wherein the top gas, consisting of unreacted hydrogen and water vapor, is cooled, dewatered in a flash unit, and partially recycled to the reactor inlet after recompression and reheating. This recycle loop improves hydrogen utilization and overall process efficiency. Material and energy balances derived from the Aspen model form the basis for techno-economic analysis. Material and energy balances derived from the Aspen model (Table 1a and Table S1: Supplementary Material) form the basis of techno-economic analysis. Detailed process flowsheets, stream data, and validation results are provided in Supplementary Section S1 (Figs. S1–S2, Tables S1). The model is validated against literature-reported hydrogen consumption (≈53.5–59.6 kg/t DRI) [52,53], with the present study yielding 55.7 kg/t DRI, confirming consistency with industrial benchmarks and shown in Table S2: Supplementary Material. For the techno-economic analysis, a more conservative design value of 59.6 kg H_2/t DRI was adopted to account for operational inefficiencies, performance degradation, and uncertainties which are not explicitly captured by the steady-state process model.

Electric heating is assumed for both hydrogen and iron ore preheating to maintain a low-carbon process boundary. The electricity requirement of the H_2-DRI process is a combination of electricity demand for hydrogen production through electrolyzer and auxiliary electricity demand associated with the process requirement such as hydrogen recycle compressor and electrical heaters. Electrolyzer electricity demand is calculated based on electrolyzer specific electricity consumption, considering alkaline electrolyzer, electrolyzer array efficiency and hydrogen demand from DRI process as per equations (1) and (2).

$$E_{DRI} = E_{ELEC} + E_{AUX} \quad (1)$$

Table 1a
Summary of key material, energy, and equipment parameters used in the Aspen Plus–based process model.

Key variables	Value	Unit	Reference
Electrolyzer array efficiency (E_{eff})	85	%	[52]
Alkaline Electrolyzer specific energy consumption (E_{SEC})	50	kWh/kg	[53]
Iron ore required for 1 MTPA DRI	1.5	MTPA	ASPEN model
Iron ore consumption for 1MTPA DRI	172	t/h	ASPEN model
H2 required for 1 MTPA DRI ($H_{2\text{-DRI}}$)	6.8	t/h	ASPEN model
Electricity required for DRI (E_{DRI})	3.55	MWh/ton of DRI	ASPEN Model Calculation
Electrolyzer required for 1 MTPA DRI	400	MW	ASPEN Model Calculation
Energy Heater Input	35	MW	ASPEN Model Calculation
Compressor capacity	2	MW	ASPEN Model Calculation

Where:

E_{DRI} (MWh/ton of DRI) = Electricity required in H_2-DRI process.

E_{ELEC} (MWh/ton of DRI) = Electricity required in electrolyzer.

E_{AUX} (MWh/ton of DRI) = Electricity required in auxiliary system (heaters + compressor)

$$E_{ELEC} = \frac{E_{SEC} * H_{2-DRI}}{E_{eff} * 1000} \quad (2)$$

Where:

E_{ELEC} (MWh/ton of DRI) = Electricity required in electrolyzer.

E_{SEC} (kWh/kg) = Electrolyzer specific energy consumption

$H_{2\text{-}DRI}$ (kg/ton of DRI) = H_2 required/ton of DRI.

E_{eff} = Electrolyzer array efficiency.

Specific energy consumption represents stack-level electrolysis performance, whereas the efficiency term reflects losses associated with the complete electrolyzer system, including balance-of-plant and array-level operation.

NG-DRI process description and modelling

The NG–DRI process is modeled using a MIDREX-type configuration in Aspen Plus under steady-state conditions to establish the fundamental model for material and energy balance [54]. A schematic representation of the NG–DRI process is provided in Fig. S3 (Supplementary Material). Natural gas (assumed primarily as methane) is reformed in a high-temperature catalytic reformer (~900 °C, Ni-based catalyst) through a combination of steam reforming, CO_2 reforming, and partial oxidation reactions to generate a reducing gas mixture composed of CO and H_2. This gas is introduced at the base of a shaft furnace, where it reacts counter-currently with descending iron ore at temperatures of approximately 900–950 °C, leading to stepwise reduction of iron oxides ($Fe_2O_3(s) \rightarrow Fe_3O_4(s) \rightarrow FeO(s) \rightarrow Fe(s)$) (Table S4: Supplementary Material). The process is represented by using stoichiometric reactor blocks for both reforming and shaft furnace reduction stages, enabling explicit simulation of reaction pathways and material flows (Fig. S4). A closed-loop configuration is implemented, wherein the top gas is cooled, dewatered, and partially recycled to the reformer after recompression to maintain gas composition and improve energy efficiency. Thermodynamic properties are evaluated using the Peng–Robinson equation of state. The simulation is initially developed under open-loop conditions to establish baseline mass and energy balances and subsequently extended to a closed-loop configuration to achieve convergence through adjustment of recycle and conversion parameters. Material and energy balances derived from the simulation (Table 1b) form the basis of techno-economic evaluation. The model is validated against literature-reported plant data Parisi et al. [55] and Hamadeh et al. [56], showing good agreement in gas composition and reduction performance (Table S5-S6: Supplementary Material). Detailed reaction systems, process flowsheets, and validation results are provided in Supplementary Section S2 (Fig. S3–S4, Tables S4–S6).

CO_2 emission intensity for DRI production routes

Table 2 presents a comparative emission study for Coal-DRI, NG-DRI, and H_2-DRI production routes under a consistent process-level boundary. The system boundary is defined on a cradle-to-gate basis up to DRI production, excluding downstream steelmaking (EAF) and upstream mining/logistics. NG-DRI and H_2-DRI are derived from process-level energy consumption calculated in this work, enabling direct estimation of associated CO_2 emissions. All emission values are normalized to DRI process-level boundary to ensure comparability across routes, despite differences in fuel sources. The emission values reported combine contribution of direct process emissions (Scope 1) and electricity-related emissions (Scope 2) under Indian grid electricity conditions. A detailed breakdown of Scope 1 and Scope 2 emissions, including normalization methodology based on process energy inputs, is provided in Table S11 in the Supplementary Material. H_2-DRI exhibits a strong dependence on the carbon intensity of electricity supply. Under low-carbon or renewable electricity conditions, emissions are significantly reduced, reflecting the absence of fossil carbon input in the reduction process. However, when electricity is sourced from the current Indian grid mix, with grid emission factor of 0.71 tCO_2/MWh [57] the emission intensity increases significantly, indicating that the decarbonization potential of hydrogen-based ironmaking is critically dependent on access to low-carbon electricity.

Table 1b
Material and energy flow in NG-DRI ASPEN simulation.

Key variable	Value	Unit	Reference
Iron ore consumption for 1MTPA DRI	1.5	MTPA	ASPEN model
Natural Gas required for 1 MTPA	0.16	MTPA	ASPEN model
Natural Gas calorific value	52	MJ/kg	
Natural Gas required for 1 MTPA	$8.2*10^6$	MMBTU/annum	Unit conversion

Table 2
Comparative process-level CO_2 emission intensity for DRI production routes.

DRI-route	Indicative Emissions (t CO_2/t-DRI)	Source basis	Reference
Coal-DRI	1.8–2.5	Literature benchmark	[38]
NG-DRI	0.45	Process model (this study)	Table S11
H_2-DRI (renewable electricity)	0.1–0.25	External literature	[39]
H_2-DRI (Indian grid electricity)	2.5	Process model (this study)	Table S11

Methodology: Economic analysis

The key economic parameters used in the techno-economic assessment of the H_2-DRI process are based on the standard cost–engineering and financial–evaluation practices. The evaluation involves estimating Capital Expenditure (CAPEX) and Operating Expenditure (OPEX), followed by applying financial assumptions to calculate indicators such as Net Present Value (NPV) and levelized cost of production (LCOP).

The total capital cost of the H_2-DRI process is calculated by considering the summation of all the costs of major equipment and associated balance-of-plant components. For electrolyzer based hydrogen production, the total CAPEX (C_T) is given by equation (3):

$$C_T = (C_E + C_{BoP} + C_C) * E_c \quad (3)$$

Where:

C_E = Unit cost of electrolyzer (US$/kW).

C_{BoP} = Unit cost of Balance of Plant (US$/kW).

C_C = Unit cost of H_2 compressor (US$/kW).

E_c = Total electrolyzer capacity (kW).

For the DRI plant, fixed capital investment is estimated using the power law scaling method [58]. Equation (4) provides the relation of plant cost with scaling factor.

$$CI = CI_{ref} \left(\frac{C}{C_{ref}}\right)^x \frac{I}{I_{ref}} \quad (4)$$

where:

CI_{ref} = Reference plant cost.

C, C_{ref} = Plant capacities.

x = Economy of scale factor (typically 0.6–0.7).

I, I_{ref} = Cost price indices for current and reference years.

To perform the calculations, the following assumptions have been applied. The Chemical Engineering Plant Cost Index (CEPCI) [59] for 2024 is 800 and for 2017 is 567. Straight line depreciation has been considered, and escalations are treated consistently with a 10% real post-tax discount rate and 20-year project life as per existing practices for preliminary estimates for process engineering projects. Table 3 supported by Table S7 in Supplementary Material provides the main parameters used for the estimation of capital and operating expenditure of H_2-DRI.

To estimate the economic feasibility of the H_2-DRI process, Net Present Value (NPV) and Levelized Cost of Production (LCOP) metrics are calculated ([60,68]) as per equations (5) and (6)

$$NPV = \sum_{t=0}^{n} \frac{Cashflow_t}{(1+r)^t} - CAPEX \quad (5)$$

$$LCOP = \frac{(CAPEX^*CRF + Annual\ OPEX)}{Annual\ Production} \quad (6)$$

Where, capital recovery factor (CRF) is used to annualize capital cost invested over the project life cycle incorporating the time value of money to give a direct comparison between one time CAPEX vs annual OPEX to evaluate the NPV, it is given by equation (7):

$$CRF = \frac{r(1+r)^n}{(1+r)^n - 1} \quad (7)$$

where r is the annualized discount rate and n is the project lifetime (in years). CRF enables conversion of total CAPEX into equivalent annualized cost.

The process model for 1 MTPA Natural Gas based Direct Reduction of Iron process was developed using ASPEN Plus v14 and the specifications have been provided in the supporting information. The outcomes of the model were used to develop an economic analysis for the NG-DRI process on similar lines. The financial model inputs are based on a detailed techno-economic analysis of the MIDREX process [68]. These inputs represent the total installed plant cost, including natural gas (NG) reforming, the shaft furnace, and other supporting plant equipment. The cost data are provided in Table 4 and Table S8 (Supplementary Material) and have been adjusted to 2024 US$ for consistency across all sources and scenarios.

The techno-economic assessment in this study is based on a simplified discounted cash flow model for a 1 MTPA plant of NG-DRI and H_2-DRI. Annual revenue is calculated as:

$$Revenue = P_{DRI}{}^*Q_{DRI}{}^*U$$

Where P_{DRI} represents the assumed market price of DRI, Q_{DRI} is the annual production capacity and U is the % utilization of the plant. The cash-flow structure includes initial capital expenditure (CAPEX) incurred in year 0 and annual operating expenditure (OPEX), with revenues generated over the plant lifetime. Net present value (NPV) is calculated using a real post-tax discount rate, assuming constant annual production and steady-state operating conditions. For comparative assessment between NG-DRI and H_2-DRI, a common DRI selling price of US$300/t was adopted, while sensitivity analyses use the base-case value of US$400/t. All NPV results are derived from this consistent cash-flow framework. The NG–DRI CAPEX is adopted as a total installed cost (TIC) from literature sources representing fully integrated MIDREX plants, whereas the H_2–DRI CAPEX is developed through a bottom-up cost breakdown to explicitly capture the contribution of hydrogen production systems and associated balance-of-plant components

Table 3
Main components for CAPEX and OPEX estimation of H_2-DRI.

Key variables	Value	Unit	Reference
AEL Electrolyzer CAPEX	390	US$/kW	[49,60]
Balance of Plant CAPEX	280	US$/kW	[61]
H_2 Compressor CAPEX	130	US$/kW	[61]
DRI Plant CAPEX	280	US$/ton of steel	[52]
H_2-DRI total CAPEX	600	US$ in million	Calculated
Discount factor	10%		[61,62]
Project life years	20		[63]
Cost price Index 2017; 2024	568; 800		[59]
Economy of scale factor	0.67		[64]
Renewable electricity cost in India	0.04	US$/kWh	[65]
Iron ore cost in India	79	US$/ton	[66]
Electrolyzer + BOP OPEX	3% of CAPEX		[52]
DRI Plant OPEX	3% of CAPEX		[52]
H_2-DRI plant total OPEX	195	US$/yr in million	Calculated
Utilization base value %	70		Base assumption
DRI base price	400	US$/metric-ton	[67]

Table 4
Main components for CAPEX and OPEX estimation for NG DRI.

Parameter	Value	Reference
DRI Capacity (Q_{DRI})	1.0 MTPA	[31]
NG required	0.16 MTPA	Calculated (Supplementary material)
NG-DRI total CAPEX	US$ 325 million	[69]
Project Life Cycle	20 years	[69]
Utilization base value % (U)	90	Base assumption
Natural gas price base value	US$7.5/MMBtu	[70]
Total OPEX	US$ 225 million/yr	Calculated (Supplementary material)
DRI base price (P_{DRI})	US$400/ton	[67]

Results and Discussion

The sensitivity analysis of Net Present Value (NPV) for 1 MTPA of H_2-DRI process highlights three major economic drivers: capital expenditure (CAPEX), operating expenditure (OPEX), and price of Direct Reduced Iron (DRI) as shown in Fig. 1. The analysis shows the impact of percentage variation in these parameters on project NPV.

Among these, DRI pricing exerts the most significant influence on project viability. A 25% increase in DRI pricing could improve the NPV by US$500 million. This means that a positive NPV is achievable when a premium price for low carbon DRI is assumed, indicating that market willingness to pay for low-carbon steel is critical for economic feasibility. While reductions in both CAPEX and OPEX improve H_2-DRI project economics, these measures alone are insufficient to make H_2-DRI profitable without providing an incentive to produce green steel.

The economic model developed in this study indicates that H_2-DRI capital expenditure is dominated by hydrogen production infrastructure, electrolyzer and balance of plants, collectively accounting for more than 50% of the total CAPEX. This finding is consistent, with previous findings as reported by Rosner et al. [7], who similarly identify the electrolyzer cost as the dominant cost contributor in steel making through H_2-DRI configuration. Therefore, technological advancements and cost reductions in electrolyzer systems are essential to enhance competitiveness.

Similarly, the study also shows that OPEX of Hydrogen production is strongly influenced by electricity tariffs, significantly contributing to total operating cost. This finding is reinforced by IEA's 2023 hydrogen based DRI cost model study [71], which highlights electricity pricing as the primary determinant of total operating cost. Therefore, electrolyzer efficiency plays a pivotal role in reducing electricity consumption, and future improvements in efficiency are expected to significantly lower

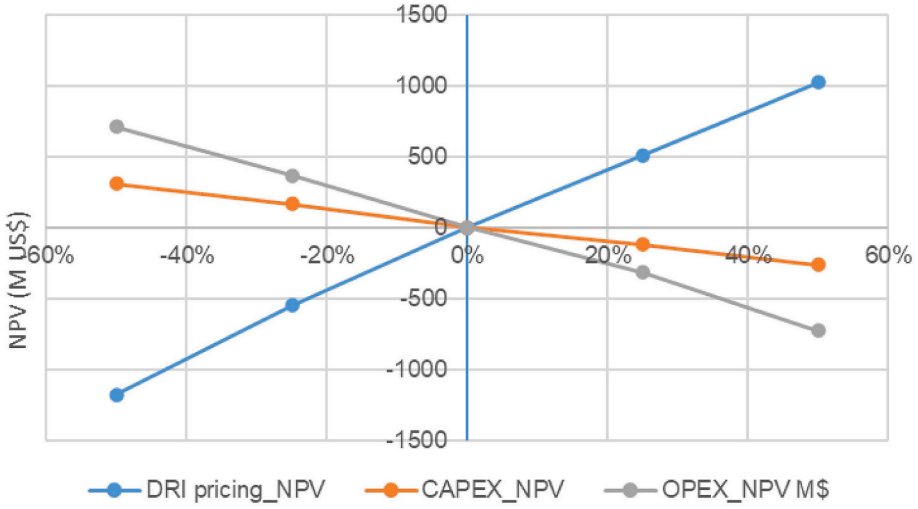


Fig. 1. NPV sensitivity analysis for the H_2-DRI process under varying DRI pricing, CAPEX, and OPEX assumptions (M$ refers to Million US$).

OPEX. Although several studies have evaluated the economics of green hydrogen production and the H_2-DRI process, this study provides a key insight. The NPV sensitivity analysis shows that project feasibility is influenced more by demand-side factors, particularly DRI selling prices, than by reductions in production costs.

Load factor sensitivity assessment

The economic viability of a 1 MTPA H_2-DRI plant in India is strongly influenced by the load factor, which represents the percentage of time the electrolyzer operates at full capacity using renewable electricity. The load factor sensitivity analysis presented in this study assumes that the H_2-DRI plant operates without dedicated hydrogen storage and grid electricity backup support through battery energy storage system (BESS). Therefore, the electrolyzer load factor is directly linked to the availability of renewable electricity supply. These boundary conditions (no storage and no grid backup) are consistently applied across both sensitivity scenarios. This assumption was adopted to directly evaluate the standalone influence of renewable electricity availability and electrolyzer utilization on H_2-DRI project economics. The renewable electricity tariff in India is relatively lower (₹3.5/kWh ~ US$0.04/kWh), in comparison to Europe (~US$0.085–0.12/kWh) [72], achieving high load factor may provide a significant competitive edge to Hydrogen based steel making in India. To assess the impact of load–factor variability on project economics, two sensitivity scenarios are examined of load factor sensitivity against CAPEX and revenue generation to assess the impact on project economics.

Sensitivity scenario 1: Fixed DRI production rate at 1 MTPA with changing CAPEX

To maintain the annual hydrogen demand for 1 MTPA of DRI production under lower load factor conditions, electrolyzer and balance of plant capacity needs to be increased to compensate for reduced operating hours. Fig. 2a is based on variations of model parameters defined in Table 3, the NPV and CAPEX sensitivity analysis for the H_2-DRI process is determined under load factor with fixed revenue.

At lower load factor (less than 60%), plant operates fewer hours per year, so the electrolyzer must be oversized to meet the hydrogen demand for 1 MTPA production. Revenue generated is insufficient to offset the high CAPEX, so NPV is steeply negative. At high load factor greater than 70%, the electrolyzer capacity approaches a minimum feasibility limit, the impact on CAPEX is minimum and consequently the impact on NPV becomes progressively smaller. This scenario is important in the design stage of the project to estimate the required capital allocation against the

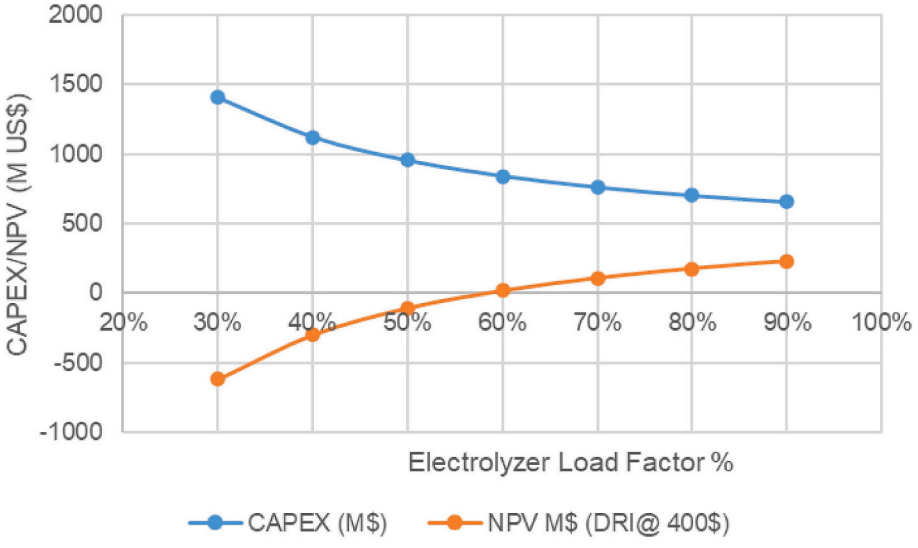


Fig. 2a. NPV and CAPEX sensitivity analysis for the H_2-DRI process under load factor with fixed revenue in M US$ (million US$).

load factor to articulate its effect on the project economics and avoid under designing of the electrolyzer capacity.

Sensitivity scenario 2: Fixed electrolyzer capacity and varying revenue

When electrolyzer capacity is fixed while load-factor varies, the analysis in Fig. 2b derived from variations of model parameters defined in Table 3, demonstrates that the reductions in load factors directly impact hydrogen output, which drives the lower DRI production and subsequently affects revenue generation. Despite the reduction in operating expenses due to operational costs, the revenue shortfall clearly dominates the economics, leading to negative NPV across the lower and mid-range load-factor conditions. This cost sensitivity emphasizes the need to ensure stable and high availability of renewable energy supply to justify high investment in green hydrogen infrastructure. These findings are valuable in providing guidance on economic penalties due to underutilization of the hydrogen DRI plant and supports investors and policy makers in addressing the challenges under condition of intermittent renewable energy supply.

Previous studies on load factor and utilization [33,73] focus more on external factors such as electricity pricing, energy source and renewable availability to understand their impact on the levelized cost of hydrogen (LCOH). In contrast, this study evaluates plant-level scenario which assesses load-factor sensitivity by examining how capital expenditure decisions are made during the design stage of the plant. Furthermore, it also provides insights on how operating expense and revenue generation during the operational phase, collectively shapes the overall techno-economic performance of H_2-DRI system. This result demonstrates for both under designing and underutilization of the H_2-DRI plant imposes a structural economic risk that must be proactively mitigated. These insights establish a need to determine a minimum viable threshold for load factor, which will be useful to protect against large capital investments and prevent economic under performance due to insufficient plant utilization.

Mitigating the risk of low load factor in H_2-DRI plants caused by intermittent renewable energy supply is essential for economic viability. Hybrid power purchase agreements (PPAs) aggregating solar and wind sources coupled with battery energy storage system (BESS), have the potential to deliver firm and consistent supply of electricity to maintain high load factor. This approach ensures round-the-clock energy supply, significantly improving the load factor and economic viability. In 2024, hybrid auctions for PPAs (solar + wind) accounted for over 50% of India's 59 GW renewable auction volumes, signaling a strong shift towards firm and dispatchable renewable energy (FDRE) solutions [74]. This also follows the global trend of combining energy produced through solar and wind in hybrid PPAs to minimize disruption of renewable energy for high industrial utilization [75].

H_2-DRI discount rate sensitivity

A sensitivity analysis of discount rate was performed to assess the impact of financing assumptions on project viability based on financial data mentioned in Table 3. The discount rate was varied between 6% and 14%, representing a realistic range of financing conditions for industrial projects in India. The results (Fig. 3) show that NPV is highly sensitive to discount rate, decreasing from US$ 258 million at 6% to US$ (−) 105 million at 14%. The project remains economically viable up to a discount rate of approximately 11%, beyond which NPV becomes negative under the base-case assumptions. In contrast, LCOP increases more gradually from US$354/t to US$409/t of DRI, indicating that financing assumptions primarily impact investment viability rather than production cost competitiveness. Lower discount rate scenarios can be interpreted as reflecting favorable financing conditions with higher debt participation, while higher discount rates represent higher-risk investment conditions, which are typical challenges for capital-intensive projects in the Indian context.

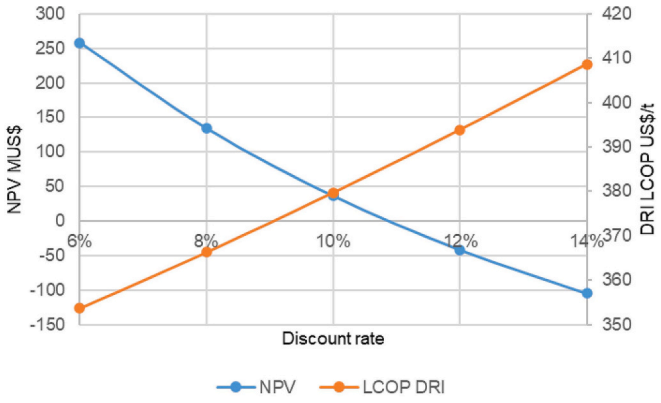


Fig. 3. Sensitivity of NPV and levelized cost of production DRI production (LCOP) to discount rate for a 1 MTPA H_2–DRI plant (70% utilization).

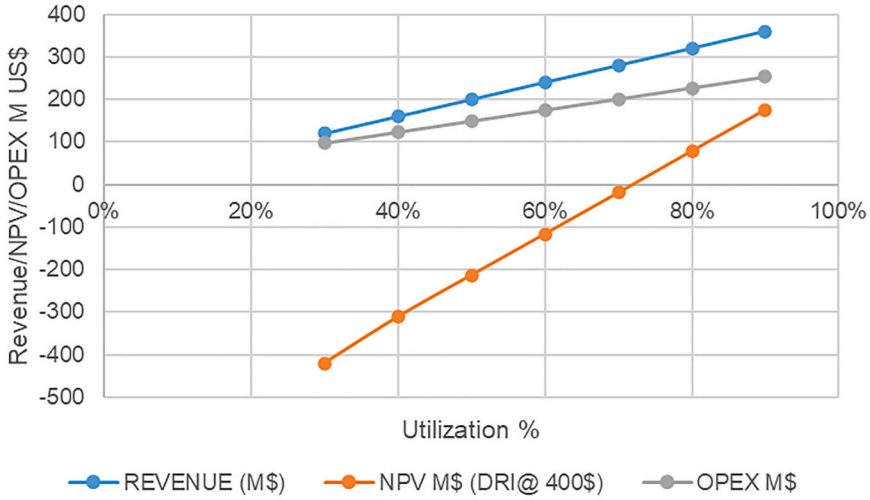


Fig. 2b. NPV CAPEX and OPEX sensitivity analysis for the H_2-DRI process under load factor with varying utilization.

H_2-DRI economic comparison against natural gas and coal-based fuel

The transition from natural gas based DRI to hydrogen based DRI represents one of the most critical technological shifts to drive decarbonization of the iron and steel industry. As India explores the decarbonization roadmap for iron and steel sector, understanding the economic implications of this transition becomes essential. Fig. 4 presents a comparative analysis of the techno-economic performance of hydrogen-based direct reduction (H_2-DRI) and natural gas-based direct reduction (NG-DRI) for a 1 MTPA steel plant under Indian market conditions. To ensure a comparison on similar lines between NG-DRI and H_2-DRI, a DRI selling price of US$300/t and a plant utilization factor of 90% were adopted for the comparative assessment. These assumptions differ from the base-case H_2-DRI values used in the sensitivity analyses and were selected to represent typical commercial operating conditions and prevailing market realities for NG-DRI plants. The rest of the calculations are derived from CAPEX/OPEX inputs in Tables 3 and 4. The comparison examines capital expenditure (CAPEX), operating expense (OPEX), Net Present Value (NPV), and Levelized Cost of Production (LCOP) calculation based on capital recovery factor estimated as 0.117.

H_2-DRI requires significantly higher capital investment compared to NG-DRI. For the 1 MTPA case, H_2-DRI CAPEX is nearly double that of NG-DRI. This difference is primarily driven by the electrolyzer system and balance-of-plant components, which accounts for roughly half of total H_2-DRI CAPEX. The balance of plant components includes electric power supply units (e.g. transformers), water purification (demineralization plant), heat exchangers for energy transfer, and gas drying and separation [76]. In contrast, NG-DRI benefits from technologically matured process and lower infrastructure costs. Operating costs for H_2-DRI are also higher than NG-DRI, with the dominant contributor to H_2-DRI OPEX is electricity cost for hydrogen production, which represents nearly half of the total operating expenses.

At the other end NG-DRI relies on natural gas pricing, which remains relatively competitive in India and global markets. Consequently, the levelized cost of DRI production (LCOP) for H_2-DRI is substantially higher than NG-DRI, reflecting the combined impact of upfront capital cost and energy intensive operating cost. While, H_2-DRI offers near-zero CO_2 emission reduction, with strong alignment with India's long-term vision, however, the current competitiveness is constrained by high electrolyzer related capital cost and renewable electricity driven OPEX.

While, NG-DRI emerged as a more economically favorable option compared to H_2-DRI under the current market condition, it is important to address that in the Indian context, coal-based DRI remains the lowest-cost production route, explaining its long standing dominance in India's steel sector despite its significantly higher CO_2 intensity, contributing to India's position as one of the largest steel-related CO_2 emitters globally. The expanded comparison shown in Fig. 5 represents a relative economic position of coal based DRI, NG-DRI against H_2-DRI across varying renewable electricity prices. The H_2-DRI and NG-DRI routes are developed using Aspen Plus process models in this study, while the coal-based DRI route is represented using literature-reported benchmark values [77–81]. This approach is adopted due to the inherent variability of small-scale coal-based rotary kiln systems in India, and the absence of standardized and publicly validated process models. Additionally, coal-DRI is included primarily as a reference baseline, for which well-established energy and emission benchmarks are available in literature and used for calculation of LCOP (Table S10: Supplementary Material). This position natural gas as a bridge technology for India's steel decarbonization, offering competitive LCOP and significantly lower carbon intensity than coal until green hydrogen technology becomes cost competitive. However, the scalability of natural gas as a fuel depends on upgrading gas infrastructure and ensuring stable industrial gas pricing (Table S9: Supplementary Material). H_2-DRI represents the long-term decarbonization pathway, delivering near-zero emissions but encounters substantial economic challenges under current conditions. Policy interventions such as carbon pricing, green steel premiums, and incentives for renewable integration and hydrogen infrastructure are expected to play a critical role in accelerating adoption.

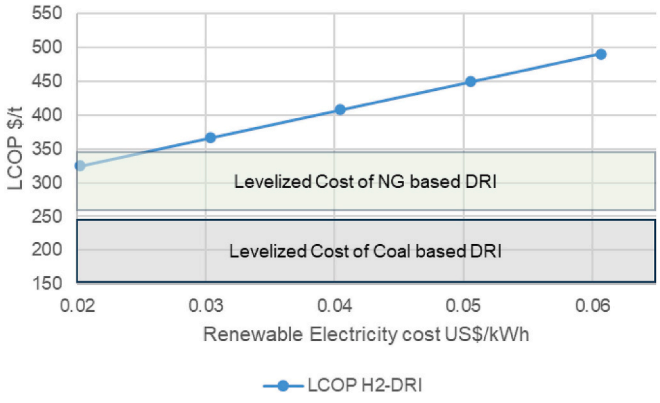


Fig. 5. LCOP comparative assessment between Coal based DRI, NG-DRI and H_2-DRI process.

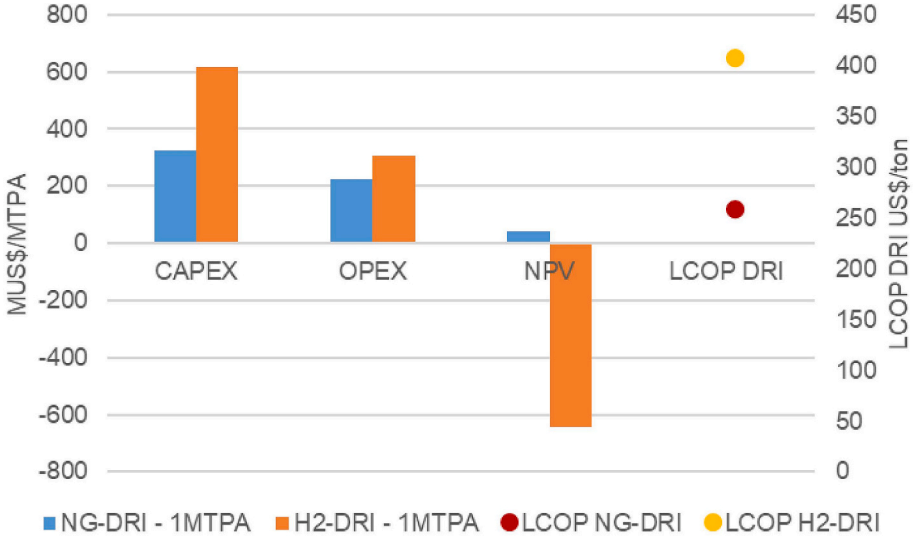


Fig. 4. NPV, CAPEX, OPEX and LCOP comparative assessment between H_2-DRI and NG-DRI process.

Building up on this foundation, the heat map in Fig. 6, derived from variation of model parameters defined in Table S7 (Supplementary Material), extends the analysis by quantifying the scale of policy in terms of carbon pricing and market support in terms of green steel premium, required to close the economic gap. A green steel premium is defined as the extra price paid for sustainably produced steel [25]. The carbon price and green steel premium are added to the revenue generation of H_2-DRI which positively impacts the NPV.

The heat map illustrates how combining the carbon pricing (US $/tCO$_2$) and green steel premium (US$/ton) impact the net present value of a 1 MTPA H_2-DRI plant in India. In the absence of external incentives in terms of carbon pricing and green steel premium, NPV is highly negative, confirming that H_2-DRI is not economically viable under current market conditions. The slope of the breakeven contour indicates that carbon pricing has a slightly stronger influence on NPV compared to green premium. The threshold value of carbon price and green steel premium required for economic competitiveness is not fixed and depends on key techno-economic assumptions, particularly OPEX (electricity cost) and CAPEX (electrolyzer cost) and Load Factor. Under the base-case assumptions used in this study, positive NPV is achieved when carbon pricing and green steel premium fall within the range of US $40–60/ton.

Therefore, the sensitivity analysis confirms that policy-driven carbon pricing and market-driven green steel premiums are strategic enablers for H_2-DRI viability. While technological improvements and renewable tariff reductions are expected to lower costs, these alone cannot close the economic viability gap. Carbon price of US$50–60/tCO$_2$ or a green premium of US$40–50/t, or a combination of both are important levers which can shift NPV into positive territory. This underscores the need for:

- Regulatory frameworks introducing carbon pricing for industrial emissions.
- Original Equipment Manufacturers (OEM) commitments to procure green steel at premium pricing.
- Integration with India's Green Hydrogen Mission [27] to align incentives and infrastructure development.

These measures, combined with falling renewable tariffs and electrolyzer cost reductions, are expected to enable H_2-DRI to become a competitive decarbonization pathway for India's steel sector by 2030.

Limitations and future work

This study presents a process-level techno-economic assessment of NG-DRI and H_2-DRI under India-specific conditions; however, several limitations remain to be addressed in future. The system boundary is restricted to direct reduced iron (DRI) production and excludes downstream steelmaking (e.g., EAF), which may influence overall economics and emissions depending on scrap availability. The analysis assumes steady-state operation without explicit hydrogen storage or grid backup, which could improve utilization under renewable intermittency. In addition, water demand for electrolysis is not explicitly modelled and may affect feasibility in water-stressed regions. The results are based on India-specific assumptions and therefore represent contextual insights, though the framework is transferable to other coal-intensive economies (e.g., China, Indonesia, South Africa). Future work should incorporate integrated DRI–EAF systems, storage and grid-backup configurations, water–energy interactions, and probabilistic approaches such as Monte Carlo simulation to capture parameter uncertainties and improve robustness of techno-economic outcomes.

Conclusions

This study developed an Aspen Plus–based techno-economic framework to evaluate the feasibility of hydrogen-based direct reduction (H_2–DRI) for India's steel sector and benchmark its performance against NG–DRI and coal-based DRI routes. The results demonstrate that H_2–DRI is a technically viable and strategically important pathway for deep decarbonization of the Indian steel industry, with the potential to significantly reduce CO_2 emissions when supported by low-carbon electricity. While current economics are influenced by higher CAPEX, electricity costs, plant utilization, and financing conditions, sensitivity analyses indicate clear pathways for improving competitiveness through technology maturation, higher renewable energy utilization, and supportive policy measures. Under the base-case assumptions, positive NPV can be achieved through a combination of carbon pricing and green steel premiums in the range of US$ 40–60/ton, although these values should be interpreted as indicative rather than prescriptive. In the near term, NG–DRI offers a practical transition pathway, delivering significantly

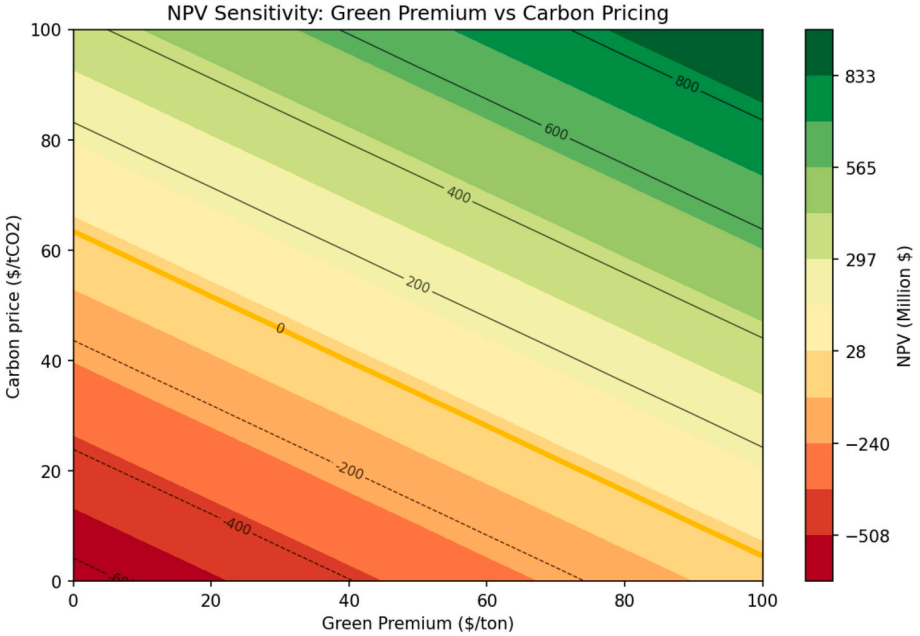


Fig. 6. Impact of carbon pricing and green steel premium on H_2-DRI economics.

lower CO_2 emissions than coal-based DRI while leveraging existing industrial infrastructure. Overall, the results suggest that coordinated policy support, continued reductions in renewable electricity and electrolyzer costs, and growing demand for low-carbon steel can position H_2–DRI as a key enabler of India's long-term green steel transition and net-zero ambitions.

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In this article, the views and opinions of the authors expressed herein do not necessarily state or reflect those of the Government of India or any agency thereof.

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The principal author (HK) has carried out his research as a part of his doctoral degree at Indian Institute of Technology Ropar and has based his analysis based on openly available data, and the calculation basis has been provided in the Supplementary Material.

Declaration of generative AI use

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CRediT authorship contribution statement

Haris Kamal: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lene Hviid:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Rutger Ijzermans:** Writing – review & editing, Supervision. **Asad H. Sahir:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Funding

This research did not receive any (specific) grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors (HK, AHS) will like to acknowledge Prof. Dhiraj K. Mahajan for introducing them to the efforts of Petroleum and Natural Gas Regulatory Board (PNGRB) in India and their reports. Also, the authors (HK, AHS) would like to acknowledge Dr. Cristina Nenu for sharing their experience and insight into this field.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.seta.2026.105278.

Data availability

All the data reported has been shared in Supplementary Material

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